CalculiX Launcher Tutorial

VERSION 0.32

http://calculix.de/
http://calculixforwin.blogspot.com/

1. Installation notes

CalculiX Launcher is the collection of portable software, that includes:
- CalculiX CCX (finite element solver);
- CalculiX CGX (pre-processor and post-processor for CCX);
- GMSH (simple parametric 3d program and mesher);
- Mesh converters from UNV, MSH, Abaqus INP, VOL formats to CCX INP format.

Most of programs are statically linked or require some common libraries (Open-GL and GTK2).
There is no special installation procedure. Xterm terminal emulator should be installed in the system (Launcher uses it by default). Some linux distributions don't have xterm, so install it as usual (in ubuntu 'sudo apt-get install xterm')

Unzip archive to any place, you want.

There is one important note, that the full path to the launcher folder should not contain spaces (blanks). For example:

/home/JohnSmith/Desktop/CL32-linux
and
/home/John_Smith/Desktop/CL32-linux
are OK,

but
/home/John Smith/Desktop/CL32-linux
is not.

(To see or copy the full path in Ubuntu with unity interface, press “Ctrl+L” in the folder)
THIS IS VERY IMPORTANT NOTE:
DON'T USE BLANKS, CYRILLIC AND ETC SYMBOLS IN THE NAMES OF FILES, FOLDERS, WHEN YOU WORK WITH LAUNCHER. FILES WON'T BE RECOGNIZED PROPERLY.

Second step is making all binaries executable. This is the standard procedure for Linux systems. At first – make executable the file, called “Launcher”, then go to the ../bin folder and do the same with all files.

Select file with mouse, then make right click and select “Properties” and go to “Permissions” tab. Allow this file to run as program (see Fig. 1) below.
Do it with all files!
Also you may cd.. to the /bin folder and type

chmod 777 *

It will mark all files as executable.

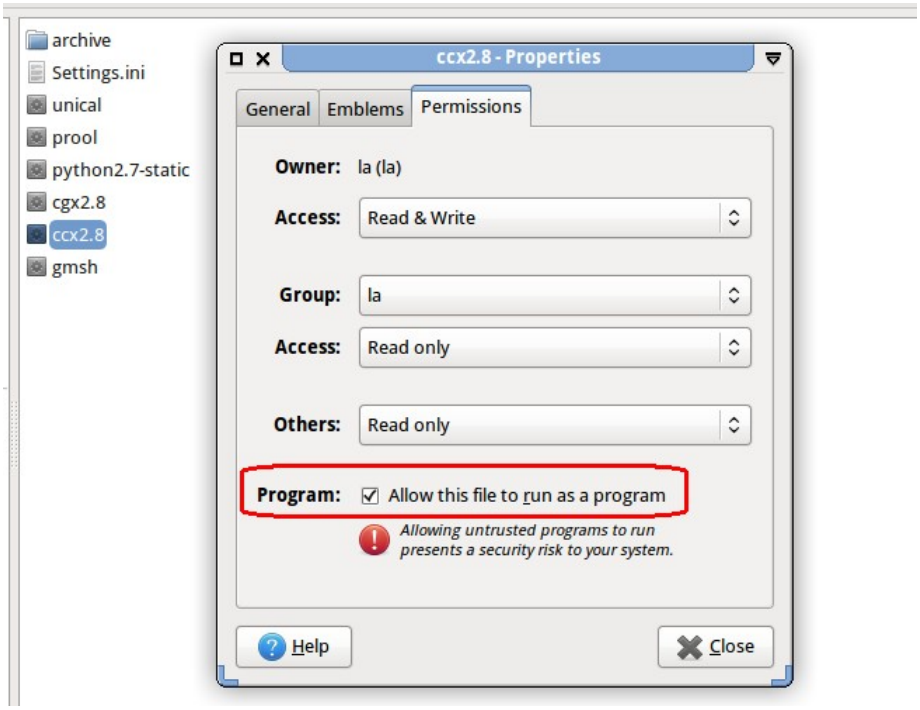


Fig. 1 – Making file executable.

Launcher comes with simple text editor for CalculiX, but you may use any other popular text and code editors, for example gedit, scite and etc. Run launcher, go to menu > Settings and fill in 4th line from above.

Default terminal emulator is x-term, but you can install and use xfce4-terminal.
In ubuntu install it with:

sudo apt-get install xfce4-terminal

Run launcher, go to menu > Settings and type 'xfce4-terminal' in 3d line from above.
Launcher-02 has two versions of solver CCX – 2.8p2 and 2.9. Some examples may work with certain version (for example truss elements in 2.9 or contact sample in 2.8 due to some changes in syntax of commands).
Most preferable configuration of hardware and operational system is 64 bit. It gives you ability to solve large problems. If you want calculation to be performed in a parallel way (for multi-cores processors), create environment variables OMP_NUM_THREADS with value equal to number of cores. To setup environment variable permanently in ubuntu, type:
sudo gedit ~/.pam_environment
and then create two lines in the text file:
OMP_NUM_THREADS=2
export OMP_NUM_THREADS
After re-log-in the calculation will run a parallel way (you will be able to see it in output for CCX).
For more information read CCX help 'How to perform CalculiX calculations in parallel'.

I will provide little bit more information about the Calculix Launcher at this page:

http://calculixforwin.blogspot.com/2015/05/calculix-launcher.html

2. General Notes About Meshers and Mesh Formats

GMSH, that is included into launcher (version 2.6) is compatible with mesh converters, provided by Dip28 user.
http://dip28p.web.fc2.com/calculix/
Newer versions are at http://geuz.org/gmsh/ may not work correctly.
There are 3 types of mesh formats can be transferred from GMSH to CalculiX:
- UNV format is supported by dip28 converter (python scripts). It works for tetrahedral and triangular mesh with nodal groups;
- Abaqus INP can be converted to CalculiX INP with dip28 script and with using Prool's converter;
- MSH format (with python script) works only for tetras.

The most powerful CAD and Mesh package is Salome-Platform.
Then you may get it at the:
http://www.code-aster.org/V2/spip.php?article303
http://www.salome-platform.org/

Read more about that and see video-lessons here:

http://calculixforwin.blogspot.com/2015/12/salome-platform-and-salome-meca.html

Output mesh format from Salome is UNV format. It works with 'unical' converter for many types of mesh. Also you may try to use combination: MED format > GMSH > CalculiX.

Netgen mesh generator (for 3d models with merged bodies) can be installed from Synaptic Package Manager. It is available in caelinux2013 distribution. Netgen have export to the abaqus INP and VOL format. VOL is supported natively in CalculiX CGX (read CGX help for that, how to deal with CAD models) and with python script, provided by dip28 (3d tetras).

In any case you may need to open converted file and correct some output manually. For example output element type for unical converter can be changed from S8R (multilayer shells for elastic and plastic analysis) to CPS8 (plane stress).
To play with mesh conversion you can use files in the
'.../samples/Converters' folder.

3. Settings for GMSH

Run GMSH from Launcher menu, go to Tools>Options>General-Advanced (see Fig. 2). Setup text editor for text file (if you want to use parametric features in future). On Fig. 2 gedit is selected as default Ubuntu editor. When you start GMSH, it creates file 'untilted.geo' for writing all your actions in the command mode. You can use this feature to make parametric models with variables. But sometimes, if you will make a mistake in the code, GMSH will crash at start. In this case you can remember the default place for 'untilted.geo' and delete (or rename it), to restore normal work.

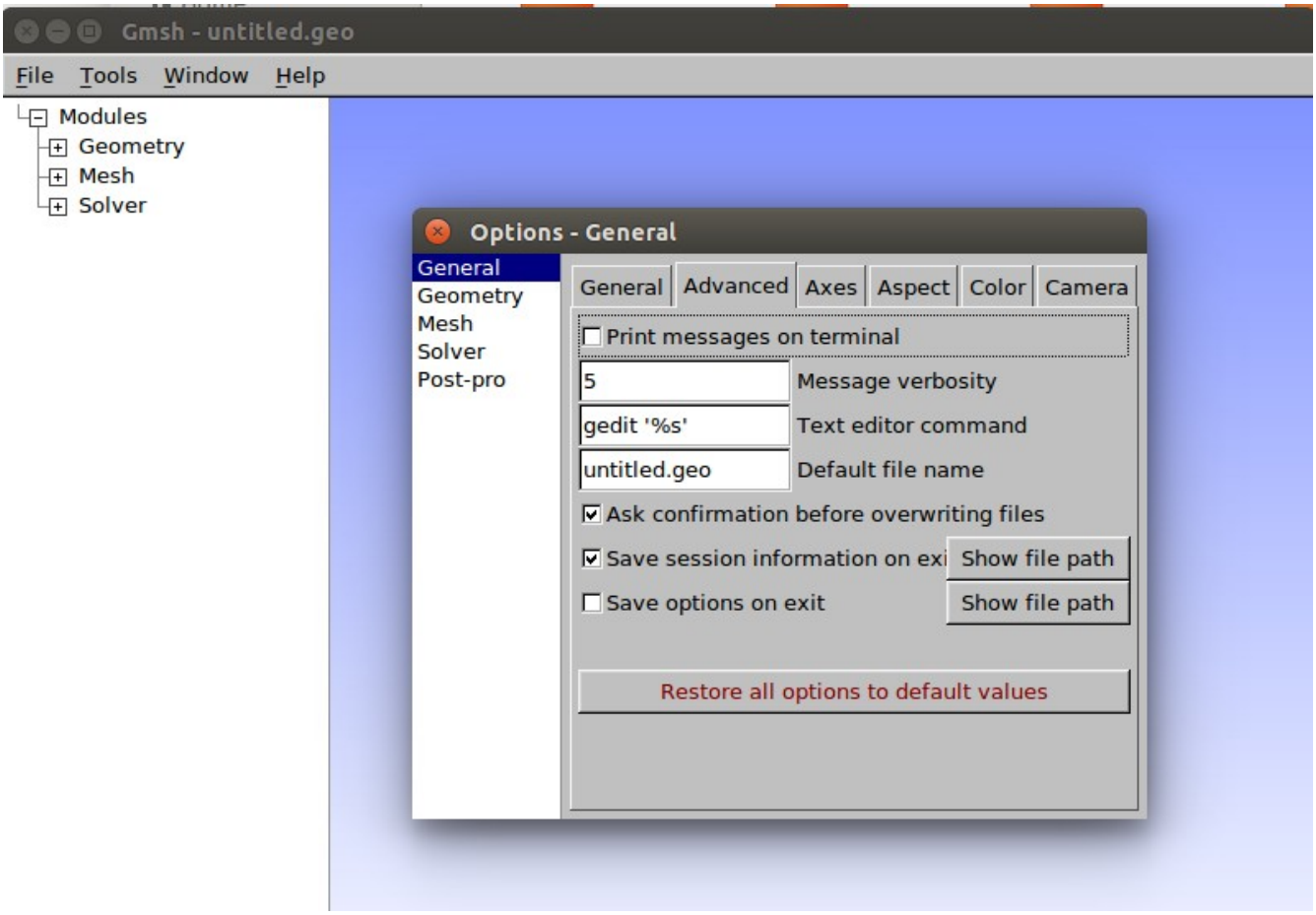


Fig. 2 – GMSH Options

4. Units in CalculiX.

CalculiX is old-school classical FEA code, and it doesn't have any certain «units» inside.
It means, that you can use any units, whatever you want, but the units should be compatible with each other.
The most simple way — to **use SI system**, just make sure that your model scaled properly and all linear dimensions of your mesh are in meters.
When you provide some input for the program, all values can be dimensionless or have some units. It is absolutely the same when you use any abstract formulas.
For example the deflection of a beam of span “L”, under the distributed load “q”, with cross section moment of inertia equal to “J” and Young's Modulus of material equal to “E” is given by the formula (bending part of full deflection).

$$y = \frac{q L^4}{8 E J}$$

There are no units that come with this kind of formulas, but you may vary with them, staying within one system for each type of value.

Let's calculate the rectangular steel beam 1x0.75 in (width=1", height=3/4"), with span

equal to 3 ft for distributed load equal to 18 lb/ft:
L = 3 ft = 3*12= 36 in;
q=18 lb/ft = $\frac{18lb}{12in}=1.5$ lb/in;
Young's Modulus E= 29,000,000 lb/in^3;
J = bh^3/12 = $\frac{1*0.75^3}{12}=0.03516$ in^4;

$$y=\frac{1.5*36^4}{8*29000000*0.03516}=0.309$$ in (or 7.8mm, or 0.0078 m)

Now let's do the same with SI system:
L=0.0254*36=0.914 m;
q=1.5 lb/in = 4.448 (N/lb)*1.5 lb /(0.0254 m) = 262.7 N/m;
E=29*10^6 lb/in^3=2*10^{11} Pa;

$$y=\frac{262.7*0.914^4}{8*2*10^{11}*1.463*10^{-8}}=0.0078$$ m

in^4 can be converted to m^4 using simple proportion:
1 in —> 0.0254 m;
1 in^4 —> (0.0254)^4 m^4 = 4.162e-7 m^4;
0.03516 in^4 => 4.162e-7*0.03516 = 1.463e-8 m^4.

When you will calculate this beam in CalculiX CCX, make sure that the mesh is properly scaled in «inches» and apply Young's modulus 29e6 psi at the *ELASTIC card.

It is little bit not so easy, when you are making modal analysis, because there is the difference in «pound-force» and «pound mass» in terms of Newton's law definitions (F=ma).
When using imperial units in CalculiX, the typical material card will be (for steel):

```
*MATERIAL,NAME=Steel
*ELASTIC
2.9000E+07,0.3
*DENSITY
7.3500E-04
```

(where the steel density 0.284 lb/in^3 is corrected by gravity acceleration =386.4 in/s^2 ; 0.284/386.4= 0.000735)

Table 1 — Input data for mild steel, depend on unit system and units for linear dimension (mesh)

Units for mesh, to be scaled	meters	inch	ft	mm
Young's Modulus (or stress units)	2e11 Pa	2.9e7 psi	4.18e9 psf	2e5 N/mm2= MPa
Poisson Ratio	0.3	0.3	0.3	0.3
Density for modal analysis	7850 kg/m3	7.35e-4 lbf/in3	15.25 lbf/ft3	7.85e-9 Ns²/mm4
Density	7850 kg/m3	0.284 lbm/in3	490.5 lbm/ft3	7.85e-9 Ns²/mm4
Gravity Acceleration	9.8 m/s2	386.4 in/s2	32.17 ft/s2	9800 mm/s2
Thermal Expansion	1.2e-5 1/degC	6.67e-6 1/degF	6.67e-6 1/degF	1.2e-5 1/degK
Thermal Conductivity	60.5 W/(m*degC)	8.09e-4 BTU/s in F	9.71e-3 BTU/s in F	60.5 N/(s*degK)
Specific Heat	434 J/(kg*degC)	0.1036 BTU/lbm*F	0.1036 BTU/lbm*F	434e6 mm²/(s²*degK)

5. How to get started

CalculiX is very efficient, fast and powerful software. It is highly recommended for those who want to learn FEA methods in details. Learning just interface of some program is not very useful, because when you will jump to other software, you will need to start from the beginning. The other benefit that when learning CalculiX you will get some skills with Abaqus INP files.
The ways to learn CalculiX are:
1) Read help files and learn examples.
2) See the video lessons: https://www.youtube.com/user/calculix09/videos
3) Solve your own problems and verification examples.
4) Learn GMSH and Salome-Platform.

To start with CalculiX CGX – see this movie: https://youtu.be/PIES0b3FHn0